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1. REVIEW

Recall that a group is a triple $(G, \cdot, e)$ consisting of a set G , a composition law $\cdot$, and an identity element e . Depending on the context, the identity may also be denoted by 1 , e_G , or 1_G .

We also defined, for each $x \in G$, the cyclic subgroup $\langle x \rangle$, which consists of all powers (positive, negative, and zero) of x . The cardinality of $\langle x \rangle$ is also called the order of x (and it can be infinite).

Let us now turn to examples.

2. EXAMPLES

Example 2.1 (The integers under addition). The triple

$$(\mathbb{Z}, +, 0) \tag{2.1}$$

is a group. Its composition law is

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n. \tag{2.2}$$

The identity element is 0 , and the inverse of m is $-m$.

Example 2.2 (The integers under multiplication: a non-example). The triple

$$(\mathbb{Z}, \times, 1) \tag{2.3}$$

is *not* a group, although it is a semigroup. Multiplication is associative and 1 is an identity, so axioms **(1)** and **(2)** hold. Axiom **(3)** fails because most integers do not have multiplicative inverses in $\mathbb{Z}$.

Example 2.3 (Integers modulo n). Let n be a positive integer. The set of residue classes modulo n is

$$\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}. \tag{2.4}$$

For example,

$$\begin{aligned} \bar{0} &= \{0, \pm n, \pm 2n, \pm 3n, \dots\}, \\ \bar{1} &= \{1, 1 \pm n, 1 \pm 2n, 1 \pm 3n, \dots\}. \end{aligned}$$

Every integer can be written $m = an + r$ with $0 \leq r < n$, and then $\bar{m} = \bar{r}$. Addition is defined by

$$\bar{m} + \bar{\ell} = \overline{m + \ell}. \tag{2.5}$$

Thus

$$(\mathbb{Z}/n\mathbb{Z}, +, \bar{0}) \quad (2.6)$$

is a group, called the *cyclic group of order n* .

In the preceding examples that are groups, the group operation is both associative and commutative.

Example 2.4 (The general linear group). The set

$$\mathrm{GL}_n(\mathbb{R}) = \{A : A \text{ is an invertible } n \times n \text{ matrix with entries in } \mathbb{R}\} \quad (2.7)$$

is a group under matrix multiplication. Its identity is the identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}. \quad (2.8)$$

Thus the group is written $(\mathrm{GL}_n(\mathbb{R}), \cdot, I_n)$, and the composition law sends (A, B) to AB . Note that the group $\mathrm{GL}_n(\mathbb{R})$ carries additional geometric structure: it is a topological space, a differentiable manifold, and a real algebraic group. In particular, it is a *Lie group*.

Example 2.5 (The special linear group). Consider the following subset of the previous example:

$$\mathrm{SL}_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) = 1\}. \quad (2.9)$$

It is a group under matrix multiplication. Indeed, $I_n \in \mathrm{SL}_n(\mathbb{R})$, and if $A, B \in \mathrm{SL}_n(\mathbb{R})$, then

$$\det(AB) = \det(A)\det(B) = 1, \quad (2.10)$$

so $AB \in \mathrm{SL}_n(\mathbb{R})$. Moreover,

$$\det(A^{-1}) = \det(A)^{-1} = 1, \quad (2.11)$$

so $A^{-1} \in \mathrm{SL}_n(\mathbb{R})$. Associativity is inherited from matrix multiplication. Thus $(\mathrm{SL}_n(\mathbb{R}), \cdot, I_n)$ is a group.

3. SUBGROUPS

Definition 3.1 (Subgroup). Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is called a *subgroup* of G if H , under the restriction of the operation on G , is itself a group.

Not all subsets are groups. For example, the subset $\{0, 3\} \subset \mathbb{Z}$ is not a group under addition, since it is not closed under addition.

Proposition 3.1. Let $(G, \cdot, e)$ be a group, and let $H \subseteq G$ be a subset satisfying the following conditions:

- (1) $e \in H$.
- (2) H is closed under the group operation: if $h_1, h_2 \in H$, then

$$h_1 \cdot h_2 \in H. \quad (3.1)$$

- (3) H is closed under inverses: if $h \in H$, then

$$h^{-1} \in H. \quad (3.2)$$

Then H is a subgroup of G .

Proof. The operation on H is the restriction of the operation on G . Associativity is therefore inherited from G , so it does not need to be checked separately. By condition (2) H is closed under this operation, and since e is an identity in G , it also satisfies the identity property of H . Finally, condition (3) ensures that the inverse in G of every element of H also belongs to H . $\square$

Remark. Suppose $H \subseteq G$ is finite. If $e \in H$ and H is closed under the group operation, then H is automatically closed under inverses. To see this, fix $h \in H$ and consider the map

$$L_h: H \longrightarrow H, \quad x \longmapsto hx. \quad (3.3)$$

This map is well-defined because H is closed under the group operation. It is injective: if $hx = hy$, then multiplying on the left by h^{-1} in G gives $x = y$. Since H is finite, L_h is therefore surjective. Because $e \in H$, there exists $y \in H$ such that $hy = e$. By uniqueness of inverses in G , we have $y = h^{-1}$. Hence $h^{-1} \in H$.

Exercise 3.1. This shows that for every $h \in H$ there exists g such that $hg = e$, so g is a right inverse. Why is g also a left inverse?